

Teorema de equivalencia del radio interno y el supremo de la derivada hiperbólica

Teorema 1. Sea $\Omega \subset \mathbb{C}$ un dominio. Entonces,

$$\frac{1}{2}R(\Omega) \leq \sup_{\substack{f \in \mathcal{H}(\mathbb{D}) \\ f(\mathbb{D}) \subset \Omega}} \left(\sup_{z \in \mathbb{D}} f^{\natural}(z) \right) \leq 2R(\Omega)$$

Demostración: La desigualdad de la derecha se sigue del teorema de Bloch invariante.

Para la desigualdad de la izquierda, tomamos $a \in \Omega$ y $r > 0$ tal que $D(a, r) \subset \Omega$.

Sea g definida en $\mathbb{D}$ por $g(z) = a + rz$. Entonces, $g \in \mathcal{H}(\mathbb{D})$ y $g(\mathbb{D}) = D(a, r) \subset \Omega$. Como $g^{\natural}(0) = (1/2) |g'(0)| = r/2$, se deduce que

$$\sup_{\substack{f \in \mathcal{H}(\mathbb{D}) \\ f(\mathbb{D}) \subset \Omega}} \left(\sup_{z \in \mathbb{D}} f^{\natural}(z) \right) \geq \frac{r}{2},$$

y como esto es cierto para cualquier disco $D(a, r) \subset \Omega$, se concluye que

$$\sup_{\substack{f \in \mathcal{H}(\mathbb{D}) \\ f(\mathbb{D}) \subset \Omega}} \left(\sup_{z \in \mathbb{D}} f^{\natural}(z) \right) \geq \frac{1}{2}R(\Omega).$$

■

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.